

MVLU COLLEGE

Aim: Python code

2a. Write a python code to create nested tuples.

```
code1.py - D:/krishan_sharma/code1.py (3.14.6)
File Edit Format Run Options Window Help
#S111 KRISHNA SHARMA

# 1. Create nested tuple with subjects
a = ("Andheri", "Ram mandir", "goregaon")
b = ("Nallasopara", "vasai", "virar")
c = ("Dahisar", "miraroad", "bhyandar")
d = (a, b, c)
print("Nested Tuple is", d)

# 2. Indexing
print("First tuple is :", d[0])
print("In First tuple first station :", d[0][0])

# 3. Negative indexing
print("The Last tuple negative index is :", d[-1])
print("Name of last tuple is :", d[-1][0])

print("\nAll tuple")

# 4. Loop through the nested tuple
for e in d:
    print(f"tuple name : {e[0]}, Code: {e[1]}")

# 5. Reverse the tuple
print("\nreverse Tuple :", d[::-1])

# 6. slice the tuple (first two station)
print("Sliced tuple (first two station) :", d[0:2])

# 7. Concatenate another subject tuple
f = ("jogeshwari", "dadar")
x = d + f
print("After Concatenation:", x)

# 8. Demonstrate immutability
try:
    d[0][1] = 200 # Trying to change subject code
except TypeError as y:
    print("\nTuple Immutability Test: Error occurred ->, y)

IDLE Shell 3.14.6
File Edit Shell Debug Options Window Help
Python 3.14.6 (tags/v3.14.6:c63a6c6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: D:/krishan_sharma/code1.py =====
Nested Tuple is (('Andheri', 'Ram mandir', 'goregaon'), ('Nallasopara', 'vasai', 'virar'), ('Dahisar', 'miraroad', 'bhyandar'))
First tuple is : ('Andheri', 'Ram mandir', 'goregaon')
In First tuple first station : Andheri
The Last tuple negative index is : ('Dahisar', 'miraroad', 'bhyandar')
Name of last tuple is : Dahisar

All tuple
tuple name :Andheri,Code:Ram mandir
tuple name :Nallasopara,Code:vasai
tuple name :Dahisar,Code:miraroad

reverse Tuple : (('Dahisar', 'miraroad', 'bhyandar'), ('Nallasopara', 'vasai', 'virar'), ('Andheri', 'Ram mandir', 'goregaon'))
Sliced tuple (first two station): (('Andheri', 'Ram mandir', 'goregaon'), ('Nallasopara', 'vasai', 'virar'))
After Concatenation: (('Andheri', 'Ram mandir', 'goregaon'), ('Nallasopara', 'vasai', 'virar'), ('Dahisar', 'miraroad', 'bhyandar'), ('jogeshwari', 'dadar'))

Tuple Immutability Test: Error occurred -> 'tuple' object does not support item assignment

>>>
```

2b. Write a python code to sort the nested tuple using sorted() function.

```
code2.py - D:/krishan_sharma/code2.py (3.14.6)
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# Original nested tuple of bikes
nested_tuple = (
    ("Honda Shine", 103),
    ("Royal Enfield Classic 350", 101),
    ("Bajaj Pulsar", 102)
)

# Sorting the nested tuple by bike code (index 1)
sorted_bikes = sorted(nested_tuple, key=lambda x: x[1])

# Display the sorted result
print("Sorted Bikes (by bike code):", sorted_bikes)

IDLE Shell 3.14.6
File Edit Shell Debug Options Window Help
Python 3.14.6 (tags/v3.14.6:c63a6c6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: D:/krishan_sharma/code2.py =====
Nested Tuple is (('Andheri', 'Ram mandir', 'goregaon'), ('Nallasopara', 'vasai', 'virar'), ('Dahisar', 'miraroad', 'bhyandar'))
First tuple is : ('Andheri', 'Ram mandir', 'goregaon')
In First tuple first station : Andheri
The Last tuple negative index is : ('Dahisar', 'miraroad', 'bhyandar')
Name of last tuple is : Dahisar

All tuple
tuple name :Andheri,Code:Ram mandir
tuple name :Nallasopara,Code:vasai
tuple name :Dahisar,Code:miraroad

reverse Tuple : (('Dahisar', 'miraroad', 'bhyandar'), ('Nallasopara', 'vasai', 'virar'), ('Andheri', 'Ram mandir', 'goregaon'))
Sliced tuple (first two station): (('Andheri', 'Ram mandir', 'goregaon'), ('Nallasopara', 'vasai', 'virar'))
After Concatenation: (('Andheri', 'Ram mandir', 'goregaon'), ('Nallasopara', 'vasai', 'virar'), ('Dahisar', 'miraroad', 'bhyandar'), ('jogeshwari', 'dadar'))

Tuple Immutability Test: Error occurred -> 'tuple' object does not support item assignment

>>>
===== RESTART: D:/krishan_sharma/code2.py =====
Sorted Bikes (by bike code): (('Royal Enfield Classic 350', 101), ('Bajaj Pulsar', 102), ('Honda Shine', 103))

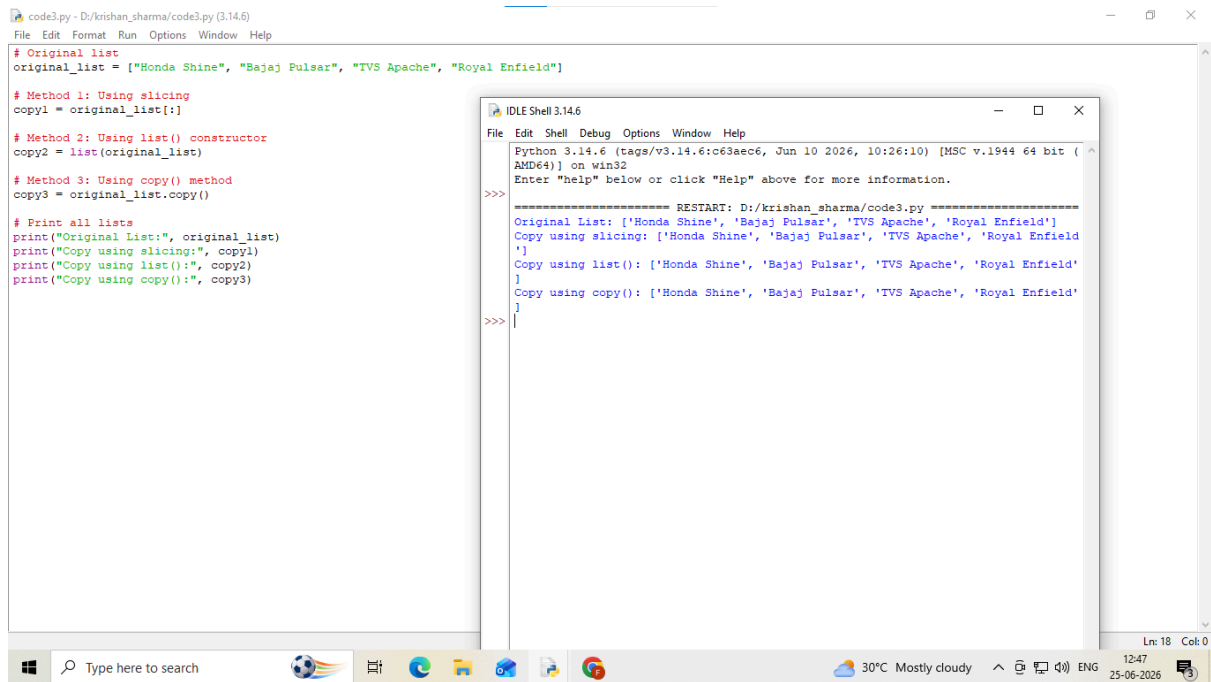
>>>
```

2c. Write a python code to copy or clone list.

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The screenshot shows a Python IDE with a file named `code3.py` and an IDLE Shell window. The code in `code3.py` defines an original list of bike models and demonstrates three methods to create a copy: slicing, the `list()` constructor, and the `copy()` method. The IDLE Shell output shows the execution results for each method, confirming that all three produce identical copies of the original list.

```
# code3.py - D:/krishan_sharma/code3.py (3.14.6)
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# Original list
original_list = ["Honda Shine", "Bajaj Pulsar", "TVS Apache", "Royal Enfield"]

# Method 1: Using slicing
copy1 = original_list[: ]

# Method 2: Using list() constructor
copy2 = list(original_list)

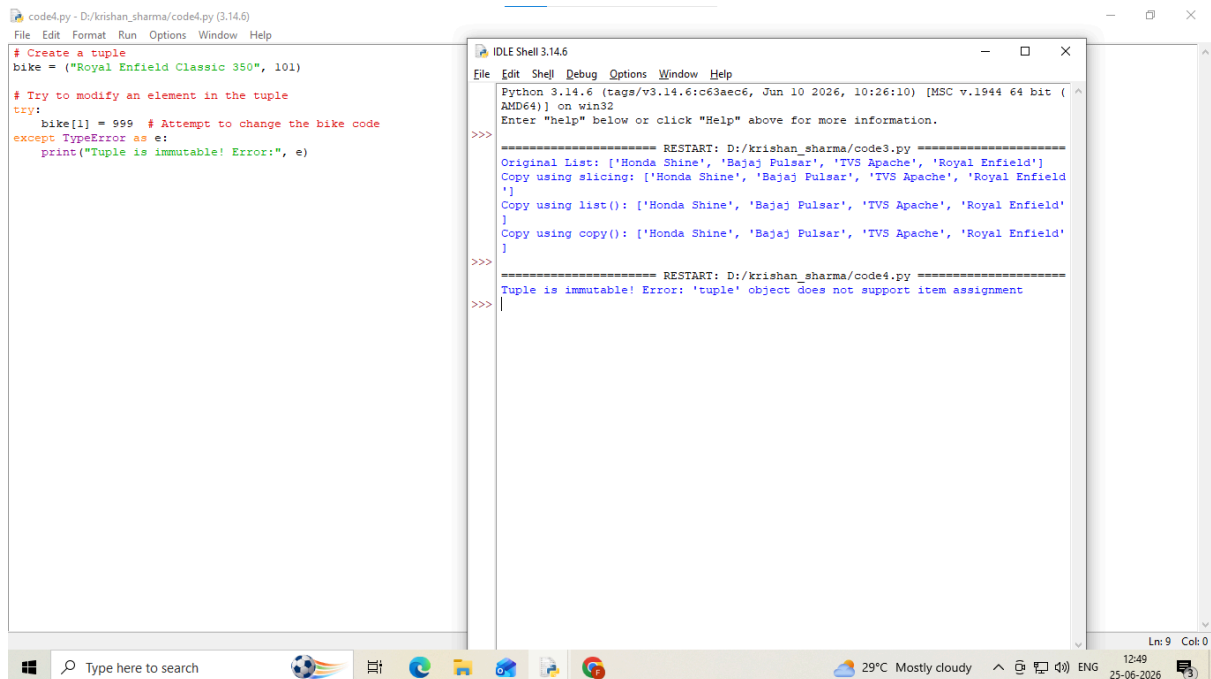
# Method 3: Using copy() method
copy3 = original_list.copy()

# Print all lists
print("Original List:", original_list)
print("Copy using slicing:", copy1)
print("Copy using list():", copy2)
print("Copy using copy():", copy3)
```

```
IDLE Shell 3.14.6
Python 3.14.6 (tags/v3.14.6:c63a6c6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: D:/krishan_sharma/code3.py =====
Original List: ['Honda Shine', 'Bajaj Pulsar', 'TVS Apache', 'Royal Enfield']
Copy using slicing: ['Honda Shine', 'Bajaj Pulsar', 'TVS Apache', 'Royal Enfield']
Copy using list(): ['Honda Shine', 'Bajaj Pulsar', 'TVS Apache', 'Royal Enfield']
Copy using copy(): ['Honda Shine', 'Bajaj Pulsar', 'TVS Apache', 'Royal Enfield']
>>>
```

2d. Write a python code to check immutability property of python tuples.



The screenshot shows a Python IDE with a file named `code4.py` and an IDLE Shell window. The code in `code4.py` creates a tuple and attempts to modify one of its elements. This results in a `TypeError`, which is caught and printed, demonstrating that tuples are immutable. The IDLE Shell output shows the error message: `'tuple' object does not support item assignment`.

```
# code4.py - D:/krishan_sharma/code4.py (3.14.6)
File Edit Format Run Options Window Help

# Create a tuple
bike = ("Royal Enfield Classic 350", 101)

# Try to modify an element in the tuple
try:
    bike[1] = 999 # Attempt to change the bike code
except TypeError as e:
    print("Tuple is immutable! Error:", e)
```

```
IDLE Shell 3.14.6
Python 3.14.6 (tags/v3.14.6:c63a6c6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: D:/krishan_sharma/code3.py =====
Original List: ['Honda Shine', 'Bajaj Pulsar', 'TVS Apache', 'Royal Enfield']
Copy using slicing: ['Honda Shine', 'Bajaj Pulsar', 'TVS Apache', 'Royal Enfield']
Copy using list(): ['Honda Shine', 'Bajaj Pulsar', 'TVS Apache', 'Royal Enfield']
Copy using copy(): ['Honda Shine', 'Bajaj Pulsar', 'TVS Apache', 'Royal Enfield']
>>>

===== RESTART: D:/krishan_sharma/code4.py =====
>>>
Tuple is immutable! Error: 'tuple' object does not support item assignment
>>>
```

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Question 1

```
codeql.py - D:/krishan_sharma/codeql.py (3.14.6)
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# a. Create a tuple with 5 different elements and print it
bikes = ("Honda Shine", "Bajaj Pulsar", "TVS Apache", "Royal Enfield", "Yamaha R15")
print("a. Tuple:", bikes)

# b. Access the first and last elements using indexing
print("\nb. First element:", bikes[0])
print("    Last element:", bikes[-1])

# c. Slice a tuple and print the middle 3 elements
print("\nc. Middle 3 elements:", bikes[1:4])

# d. Concatenate two tuples and print the result
more_bikes = ("KTM Duke", "Hero Splendor")
#Sill KRISHNA SHARMA
combined_bikes = bikes + more_bikes
print("\nd. Concatenated Tuple:", combined_bikes)

# e. Reverse a tuple using slicing
print("\ne. Reversed Tuple:", bikes[::-1])

# f. Count how many times an element appears in a tuple
numbers = (10, 20, 30, 20, 40, 20)
print("\nf. Count of 20:", numbers.count(20))

# g. Find the index of a specific element in a tuple
print("\ng. Index of 'TVS Apache':", bikes.index("TVS Apache"))

# h. Check if an element exists in a tuple
element = "Yamaha R15"
if element in bikes:
    print("\nh. ", element, "exists in the tuple.")
else:
    print("\nh. ", element, "does not exist in the tuple.")

# i. Convert a list to a tuple
bike_list = ["KTM Duke", "Hero Splendor", "Suzuki Gixxer"]
bike_tuple = tuple(bike_list)
print("\ni. Converted Tuple:", bike_tuple)

# j. Sort a tuple of numbers in ascending order

IDLE Shell 3.14.6
File Edit Shell Debug Options Window Help

Python 3.14.6 (tags/v3.14.6:c63a6c6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: D:/krishan_sharma/codeql.py =====
a. Tuple: ('Honda Shine', 'Bajaj Pulsar', 'TVS Apache', 'Royal Enfield', 'Yamaha R15')
b. First element: Honda Shine
   Last element: Yamaha R15
c. Middle 3 elements: ('Bajaj Pulsar', 'TVS Apache', 'Royal Enfield')
d. Concatenated Tuple: ('Honda Shine', 'Bajaj Pulsar', 'TVS Apache', 'Royal Enfield', 'Yamaha R15', 'KTM Duke', 'Hero Splendor')
e. Reversed Tuple: ('Yamaha R15', 'Royal Enfield', 'TVS Apache', 'Bajaj Pulsar', 'Honda Shine')
f. Count of 20: 3
g. Index of 'TVS Apache': 2
h. Yamaha R15 exists in the tuple.
i. Converted Tuple: ('KTM Duke', 'Hero Splendor', 'Suzuki Gixxer')
j. Sorted Tuple: (10, 20, 30, 40, 50)
k. Repeated Tuple: ('Bike', 'Bike', 'Bike')
l. Tuple is immutable! Error: 'tuple' object does not support item assignment

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25-06-2026
```

```
print("\nh. ", element, "does not exist in the tuple.")

# i. Convert a list to a tuple
bike_list = ["KTM Duke", "Hero Splendor", "Suzuki Gixxer"]
bike_tuple = tuple(bike_list)
print("\ni. Converted Tuple:", bike_tuple)

# j. Sort a tuple of numbers in ascending order
num_tuple = (50, 10, 40, 20, 30)
sorted_tuple = tuple(sorted(num_tuple))
print("\nj. Sorted Tuple:", sorted_tuple)

# k. Repeat a tuple 3 times using the * operator
print("\nk. Repeated Tuple:", ("Bike",) * 3)

# l. Check the immutability property of tuples
bike = ("Royal Enfield", 101)

try:
    bike[1] = 999
except TypeError as e:
    print("\nl. Tuple is immutable! Error:", e)

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12:53
25-06-2026
```

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Question 2:

The screenshot shows a Windows desktop with two windows. The left window is a code editor titled 'CODEQ2.PY - D:/krishan_sharma/CODEQ2.PY (3.14.6)' containing a Python script. The right window is the 'IDLE Shell 3.14.6' showing the execution output of the script.

```
# a. Find the largest number in a list
numbers = [10, 25, 8, 45, 30]
print("a. Largest number:", max(numbers))

# b. Remove duplicates from a list
dup_list = [10, 20, 10, 30, 20, 40]
unique_list = list(set(dup_list))
print("\nb. List after removing duplicates:", unique_list)

# c. Count how many even numbers are in a list
num_list = [1, 2, 3, 4, 5, 6, 8]
even_count = 0

for num in num_list:
    if num % 2 == 0:
        even_count += 1

print("\nc. Number of even numbers:", even_count)

# d. Input 5 numbers and store them in a list
input_list = []

for i in range(5):
    num = int(input("Enter a number: "))
    input_list.append(num)

print("\nd. List entered by user:", input_list)

# e. Function to return the average of all numbers in a list
def average(lst):
    return sum(lst) / len(lst)

avg = average(numbers)
print("\ne. Average of the list:", avg)

# f. Convert a string into a list of characters using list()
text = "Python"
char_list = list(text)
print("\nf. List of characters:", char_list)

# g. Join all elements of a list into a single string using join()
```

The IDLE Shell output shows the following results:

```
Python 3.14.6 (tags/v3.14.6:c63a6c6, Jun 10 2026, 10:26:10) [MSC v.1914 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

===== RESTART: D:/krishan_sharma/CODEQ2.PY =====
a. Largest number: 45

b. List after removing duplicates: [40, 10, 20, 30]

c. Number of even numbers: 4
Enter a number: 20
Enter a number: 21
Enter a number: 23
Enter a number: 35
Enter a number: 54

d. List entered by user: [20, 21, 23, 35, 54]

e. Average of the list: 23.6

f. List of characters: ['P', 'y', 't', 'h', 'o', 'n']

g. Joined String: Python for Data Science

>>>
```

The screenshot shows a Windows desktop with a code editor window titled 'CODEQ2.PY - D:/krishan_sharma/CODEQ2.PY (3.14.6)'. The code editor contains a Python script, which is a continuation of the script from the previous screenshot.

```
# e. Function to return the average of all numbers in a list
def average(lst):
    return sum(lst) / len(lst)

avg = average(numbers)
print("\ne. Average of the list:", avg)

# f. Convert a string into a list of characters using list()
text = "Python"
char_list = list(text)
print("\nf. List of characters:", char_list)

# g. Join all elements of a list into a single string using join()
words = ["Python", "for", "Data", "Science"]
joined_string = " ".join(words)
print("\ng. Joined String:", joined_string)
```

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